

The anchor tag — <a>

```
<a href="https://khanacademy.org">
  Practice math on Khan Academy
</a>
```

Three parts: the tag, the `href` attribute, and the text between. The text is what the user clicks and reads.

Anchor is the name of the tag. **Link** is what it makes. The two words get used interchangeably, but the tag itself is called an anchor.

`href` stands for **H**ypertext **RE**ference. It tells the browser where the link goes.

Attributes go in the opening tag only. Never `</a href="...">`. This is a general rule for every HTML tag you will ever write.

Four destinations a link can point to

DESTINATION	EXAMPLE	NOTES
Another website	<code>href="https://www.nasa.gov"</code>	Must start with <code>https://</code> .
Another page of yours	<code>href="about.html"</code>	Same folder as the current page.
A section of this page	<code>href="#contact"</code>	Needs a matching <code>id</code> on the target element.
An email or phone	<code>href="mailto:you@site.com"</code>	Opens the user's mail or phone app.

Absolute vs relative URLs

Absolute — full mailing address

```
<a href="https://www.w3schools.com">
  Tutorials
</a>
```

- Starts with `https://`
- Works from anywhere
- Use for **other people's** websites

Relative — directions from here

```
<a href="about.html">
  About Me
</a>
```

- No `https://`, no domain
- "A file sitting next to this one"
- Use for **your own** pages

The mistake everyone makes. `<a href="www.google.com">Google</a>` looks correct but fails. With no `https://` at the front, the browser thinks `www.google.com` is a *file* sitting next to your page and shows a "page not found" error on a site you know exists.

Opening a link in a new tab

```
<a href="https://nasa.gov"
  target="_blank">
```

USE `TARGET="_BLANK"`
FOR

Links leaving your site

☒ Yes

```
Explore NASA missions (opens in a new tab)
</a>
```

Note the underscore: `_blank`, not `blank`.

Links to your own pages **✗ No**

Opening your own pages in new tabs buries the user in tabs and breaks the back button. Reserve the new tab for links that leave you forever.

Accessibility note. A sighted user sees a new tab appear. A screen reader user just hears the page change. Say so in the link text: `...(opens in a new tab)`.

Link text — the part most people get wrong

✗ Weak link text

```
<p>For math, <a href="...">click here</a>.</p>
<p>For science, <a href="...">click here</a>.</p>
<p>For history, <a href="...">click here</a>.</p>
```

Reads fine in a paragraph. Fails everywhere else.

✓ Strong link text

```
Practice math on Khan Academy
Download the field trip form
Watch the NASA launch livestream
```

Test: read only the link text. Do you know where it goes?

Why it matters. A screen reader can list every link on a page as a standalone list. Out of context, the user hears “click here · click here · click here” and learns nothing. The link text is the *only* thing they get.

Same-page anchors — id and

Step 1 — name the destination

```
<h2 id="contact">Contact Information</h2>
```

The `id` changes nothing visually. You just gave that heading a name.

Step 2 — link to it with

```
<a href="#contact">Jump to Contact</a>
```

The `#` means “look on *this* page for that id.”

Three rules, three traps.

- The `#` goes in the `href`, *never* in the `id` itself. `id="#contact"` is wrong.
- An `id` is unique. One `id="contact"` per page, never two — only the first will be found.
- No spaces in an `id`. Use `my-hobbies`, not `my hobbies`.

The “back to top” pattern

```
<h1 id="top">My Website</h1>

<!-- much later on the page -->
<a href="#top">Back to top</a>
```

Put an `id` on the top heading of the page and you can always send the user back home with a single link. Every long page on the web has one.

Email and phone links

Email

```
<a href="mailto:you@site.com">
  Email me
</a>
```

No //, no https — just `mailto:` then the address.

```
<a href="mailto:you@site.com?subject=Homework">
  Homework question
</a>
```

`?subject=` pre-fills the subject line.

Phone

```
<a href="tel:+12015551234">
  Call the main office
</a>
```

Digits only. Letters like `555-TRIP` will not dial.

`mailto:` only works if the user has an email app set up. `tel:` does nothing on most desktop computers. Show the raw address or number as text too, so the link is useful either way.

Common mistakes to avoid

MISTAKE	WHAT GOES WRONG
Missing <code>https://</code>	<code>href="www.google.com"</code> looks for a file called <code>www.google.com</code> next to your page.
Missing <code>#</code>	<code>href="contact"</code> is not an anchor link. The browser hunts for a file called <code>contact</code> .
Duplicate <code>id</code>	Two elements with the same name. Only the first is found, and the second link silently does nothing.
"Click here"	Useless when the link text is read out of context, as screen readers do.
Underscore missing in <code>target</code>	<code>target="blank"</code> opens a tab <i>named</i> blank, not a new tab. It has to be <code>_blank</code> .

Today's syntax — quick reference

GOAL	CODE
External site	<code>Text</code>
New tab	<code>target="_blank"</code>
Your own page	<code>Text</code>
Section of this page	<code>Text</code>
Name a section	<code><h2 id="name">Heading</h2></code>
Email	<code></code>
Phone	<code></code>

Before you submit — test every link

- Click each one. Does it land where you meant?
- Do external links open in a new tab?
- Do the `#` links jump to the right section?
- Do the email and phone links point to the correct address and number?

A link that looks right and goes nowhere still counts as broken.

What is next. Your pages can now point anywhere. Lesson 5 groups content with lists — and a navigation menu is nothing more than a list of the links you built today. Yesterday's lesson becomes today's building block.

Keep this sheet. Every URL you write from now on is either absolute or relative, and this sheet is the tie-breaker when you are not sure.